

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0404

Brownfield Switching 138 kV
June 25, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0404. The Interconnection Customer proposes a 750.6 MW natural gas combined cycle facility interconnecting at the Brownfield Switching 138 kV substation in the West Virginia area. The request was received on November 19, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 206 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$66,484,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lindon / Eagle Vly In Substation 138 KV	Eagle Vly In Substation / Marsh Run Ct Substation 138 KV	256.6	176	38.7
Lindon / Pit 4 138 KV line	Eagle Vly In / Marsh Run Ct 138 KV	240.6	186	38.7
Marsh Run Ct / Eagle Valley In 138 KV	Lindon / Eagle Vly In 138 KV line	205.5	250	45.2
Garces / Clendenin 138 KV Line	Eagle Valley In / Brownfield Switching 138 KV L	203.3	254	36.5
Eagle Vly In / Brownfield Switching 138 KV	Capitol Hill / Chesterfield Ave 138 KV	201.9	374	61.3
Marsh Run Ct / Malad 138 KV Line	Eagle Vly In / Lindon 138 KV	199.9	254	45.2
Brownfield Switching / Garces 138 KV	Eagle Vly In / Brownfield Switching 138 KV	192.4	384	100.0
Panda / Menlo 138 KV	Fishkill Plains Substation / Dillard-Orange 345 KV Line	145.5	229	5.6
Hampshire Paper / Solana 138 KV	Fishkill Plains / Dillard-Orange 345 KV	142.9	245	5.6
Farragut / Wards Hollow 138 KV	Brownfield Switching / Eagle Valley In 138 KV	134.7	178	13.4
Clinton Atle / Lodi Stig 138 KV	Brownfield Switching / Eagle Vly In 138 KV	127.0	198	13.4
Slana / Northern Steel Casting Substation 138 KV	Fishkill Plains / Dillard-Orange 345 KV	125.6	275	5.6
Clinton Atle / Farragut 138 KV	Brownfield Switching / Eagle Vly In 138 KV	118.7	204	13.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Laurelwood / Clendenin 138 KV	Brownfield Switching / Eagle Vly In 138 KV line	114.6	226	20.2
Lodi Stig / Chesterfield Ave 138 KV	Brownfield Switching / Eagle Valley In 138 KV	114.3	361	21.4
Wards Hollow / Menaul 138 KV	Eagle Valley In / Brownfield Switching 138 KV	111.9	204	13.4
Clendenin Substation / Sumpter Substation 138 KV	Brownfield Switching / Eagle Valley In 138 KV	111.5	226	16.3
Surrey Ridge / Sumpter 138 KV	Brownfield Switching / Eagle Vly In 138 KV	110.9	223	16.3
Fishkill Plains / Middle Fork 345 KV	Simpson Stuart Tap / Fishkill Plains 345 KV	108.4	1,241	8.7
Chemical / Washington St 138 KV Line	Brownfield Switching / Eagle Vly In 138 KV	107.2	509	42.1
Washington St / Capitol Hill Substation 138 KV Line	Brownfield Switching / Eagle Vly In Substation 138 KV	107.0	520	42.1
Chesterfield Ave / Capitol Hill 138 KV	Brownfield Switching / Eagle Valley In 138 KV Line	106.9	529	42.1
Ulmerton Transfer Center / Middle Fork Substation 345 KV L	Medora Substation / P.T.S 345 KV	106.2	1,310	6.6
Pit 4 / Forest 138 KV	Eagle Valley In / Marsh Run Ct 138 KV	105.8	258	28.7
Dillard-Orange Substation / Fishkill Plains 345 KV	Flatlik / International Paper 500 KV	104.9	2,346	14.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lindon / Eagle Vly In 138 KV	Rebuild line with higher-capacity conductor	\$2,811,000	59% (\$1,659,000)
Lindon / Pit 4 138 KV line	Rebuild line with higher-capacity conductor	\$6,964,000	59% (\$4,110,000)
Marsh Run Ct / Eagle Valley In 138 KV	Rebuild line with higher-capacity conductor	\$5,482,000	59% (\$3,243,000)
Garces / Clendenin 138 KV	Rebuild line with higher-capacity conductor	\$17,361,000	54% (\$9,305,000)
Eagle Vly In / Brownfield Switching 138 KV	Rebuild line with higher-capacity conductor	\$5,021,000	55% (\$2,759,000)
Marsh Run Ct / Malad 138 KV	Rebuild line with higher-capacity conductor	\$2,124,000	59% (\$1,256,000)
Brownfield Switching / Garces 138 KV	Rebuild line with higher-capacity conductor	\$1,850,000	100% (\$1,850,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Menlo / Panda 138 KV	Rebuild line with higher-capacity conductor	\$6,327,000	46% (\$2,894,000)
Solana / Hampshire Paper Substation 138 KV	Rebuild line with higher-capacity conductor	\$7,080,000	46% (\$3,238,000)
Farragut / Wards Hollow 138 KV	Rebuild line with higher-capacity conductor	\$3,528,000	45% (\$1,592,000)
Clinton Atle / Lodi Stig 138 KV	Rebuild line with higher-capacity conductor	\$1,926,000	45% (\$869,000)
Solna / Northern Steel Casting 138 KV	Rebuild line with higher-capacity conductor	\$10,421,000	46% (\$4,766,000)
Farrat / Clinton Atle Substation 138 KV line	Rebuild line with higher-capacity conductor	\$4,780,000	45% (\$2,157,000)
Laurelwood / Clendenin Substation 138 KV	Rebuild line with higher-capacity conductor	\$7,646,000	56% (\$4,316,000)
Lodi Stig / Chesterfield Ave 138 KV	Rebuild line with higher-capacity conductor	\$3,211,000	44% (\$1,408,000)
Wards Hollow / Menaul 138 KV	Rebuild line with higher-capacity conductor	\$6,576,000	45% (\$2,968,000)
Clendenin Substation / Sumpter Substation 138 KV	Rebuild line with higher-capacity conductor	\$8,735,000	55% (\$4,797,000)
Surrey Ridge / Sumpter 138 KV	Rebuild line with higher-capacity conductor	\$11,153,000	55% (\$6,124,000)
Middle Fork / Fishkill Plains 345 KV	Reconductor line	\$21,535,000	7% (\$1,551,000)
Washington St / Chemical 138 KV	Reconductor line	\$966,000	44% (\$423,000)
Capitol Hill / Washington St 138 KV	Reconductor line	\$924,000	44% (\$404,000)
Capitol Hill / Chesterfield Ave 138 KV	Reconductor line	\$336,000	44% (\$147,000)
Middle Fork / Ulmerton Transfer Ctr Substation 345 KV	Reconductor line	\$15,368,000	9% (\$1,334,000)
Pit 4 / Forest 138 KV	Reconductor line	\$678,000	62% (\$419,000)
Fishkill Plains / Dillard-Orange 345 KV Line	Reconductor line	\$41,044,000	7% (\$2,895,000)
	Total allocated to project		\$66,484,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.